

HUG: A Heterogeneous Unified Graph Framework for Conversion Rate Prediction

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Abstract

Conversion rate prediction in short-video platforms is challenged by the extreme sparsity of engagement signals and the inherent exposure bias of production recommendation policies. We present The Heterogeneous Unified Graph (HUG) framework that unifies diverse behavioral signals into a high-fidelity relational substrate. We construct a heterogeneous knowledge graph over the KuaiRand dataset, integrating five entity types (users, videos, sessions, authors, and categories) linked by twelve distinct interaction edges ranging from positive endorsements (likes, follows) to negative feedback (hate signals). To effectively capture both long-term preferences and immediate intent, we propose a dual-view decomposition architecture. A relational graph convolutional network operates over a structural subgraph to extract global topological features, while a gated graph neural network processes a sequential subgraph to model within-session dynamics. These representations are fused via a relation-aware cross-attention alignment module. Evaluations on the KuaiRand-1K benchmark demonstrate that HUG achieves an AUC-ROC of 0.7683 and an average precision of 0.6578, representing a 9.62% relative improvement in AUC and a 20.9% improvement in AP over the best sequence-based baseline Behaviour Sequence Transformer, and a 11.6% improvement in AUC over the graph-based Knowledge Graph Attention Network. baseline.

CCS Concepts

• Information systems → Recommender systems; • Computing methodologies → Machine learning approaches; Learning from implicit feedback; Neural networks; Learning latent representations; Causal reasoning and diagnostics.

Keywords

Conversion Rate Prediction, Heterogeneous Knowledge Graphs, Short-Video Recommendation, Graph Neural Networks

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1 Introduction

Conversion Rate (CVR) prediction (estimating the probability that an exposed item produces a downstream engagement action) is one of the most commercially consequential tasks in modern recommender systems. However, CVR prediction remains fundamentally

challenged by two structural properties: *sample selection bias* and *data sparsity* [8]. While existing literature has addressed these via multi-task learning or counterfactual estimators, these approaches often treat them in isolation, ignoring the rich, multi-typed relational structure that connects items to their metadata [1, 15].

To address these challenges, we propose **HUG**, a framework built on the *principled unification* of disparate data modalities. We argue that while metadata and behavior can be represented in a single graph, they possess fundamentally different temporal and structural characteristics: static topological features (who the users and authors are) and dynamic sequential intent (what the user is doing now). Processing them through a monolithic encoder often leads to sub-optimal signal conflation. Our framework provides a modular architectural progression for effectively integrating these signals:

- **Unified Relational Substrate (HUG-Unified):** We formalize a heterogeneous knowledge graph (HKG) integrating five entity types and twelve interaction edges. We demonstrate that a unified encoder, instantiated via a Heterogeneous Graph Transformer (HGT) [3] over this relational structure, provides a superior foundation for CVR compared to flat interaction sequences.
- **Decomposed Modality Encoding (HUG-Dual):** We propose a dual-encoder architecture instantiated via two Graph Neural Networks (GNNs) that decouple the HKG into structural and sequential subgraph views. By pairing a Relational Graph Convolutional Network (R-GCN) [9] for long-term topological patterns with a Gated Graph Neural Network (GGNN) [6] for short-term dynamics, we match the inductive bias of each encoder to the temporal character of its modality.
- **Cross-Attention Module:** We introduce a dual cross-attention mechanism that fuses the structural and sequential views into a unified representation. Two bidirectional attention operations run in parallel: one in which the structural video embedding attends to the session embedding, and one in which the sequential video embedding attends to the structural video embedding. This bidirectional design ensures that neither view dominates by default. The module learns, conditioned on the specific user-item-session triple, how much each signal should contribute to the final representation. A user-session alignment MLP operates in parallel to blend long-term preference with immediate session intent. The outputs of all three subcomponents are concatenated and passed through a fusion MLP with LayerNorm to produce a single fused vector $z \in \mathbb{R}^d$. A Knowledge Graph (KG) alignment gate optionally grounds the attention logits in relation path features from the HKG, providing a structural guide when relevant paths exist.

1.1 Dataset Considerations

To evaluate the effectiveness of **HUG**, we utilize the **KuaiRand-1K** dataset [2]. This dataset provides a robust environment that



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preserves the high-density relational structure required for our analysis.

KuaiRand-1K consists of 11.7 million interactions and encompasses approximately 4.37 million unique videos. A defining characteristic of this dataset is its inclusion of a random-policy intervention subset. Unlike standard production logs, which are subject to exposure bias from existing ranking algorithms, this subset consists of 43,028 interactions where videos were shown to users via a uniform random policy.

Furthermore, KuaiRand-1K provides a rich, multi-modal relational structure. It encompasses 12 distinct interaction types, ranging from strong positive signals to explicit negative feedback. This granularity is essential for constructing our HKG, as it allows the relation-aware cross-attention module to distinguish between casual browsing and high-intent engagement. The dataset features high-fidelity metadata, including user profiles, video properties, and author information. In addition to the static structural profiling data, the dataset provides chronological sequences of user actions, enabling our framework to model evolving behavioral patterns alongside fixed entity attributes. The metadata and interaction types are specified in Table 1.

Table 1: KuaiRand-1K Feature and Interaction Registry

Entity	Attributes
User Profile	age, gender, city, residence, device
Video Prop.	tags, categories, video_id, upload_type
Author Info	author_id, gender, city, type
Category	Interaction Types (12)
Positive	is_click, is_like, is_follow, is_comment, is_forward, long_view, is_profile
Negative	is_hate, is_negative
Numerical	play_time_ms, duration_ms, watch_ratio

2 Related Work

2.1 CVR Prediction and Data Sparsity

Modeling conversion behavior in recommendation systems is fundamentally challenged by the sparsity of positive signals and the mismatch between observed interactions and downstream outcomes. The Entire Space Multi-task Model (ESMM) [8] and variants such as The Entire Space Counterfactual Multi-task Model (ESCM²) [11] address this issue by jointly modeling upstream engagement signals (e.g., clicks) alongside downstream conversion behavior, enabling learning over a broader interaction space.

More broadly, recent work has emphasized the importance of reliable training and evaluation regimes when interaction data is heavily skewed or policy-driven. Datasets that incorporate randomized exposure subsets, such as KuaiRand [2], provide a more stable foundation for benchmarking recommendation models by reducing feedback loops introduced by production policies. In this work, we adopt this setting to ensure that model comparisons reflect representational capacity and generalization behavior rather than artifacts of exposure bias.

2.2 Sequential Interest Modeling

Capturing evolving user intent has historically relied on sequential architectures. Foundational models such as the Deep Interest Network (DIN) [15] use local activation units to attend to relevant historical items, while the Behavior Sequence Transformer (BST) [1] leverages self-attention to model long-range temporal dependencies. These methods have demonstrated strong performance in modeling behavioral dynamics, particularly when user preferences shift over time.

However, sequence-based approaches typically operate over item identifiers and interaction histories alone, limiting their ability to incorporate structured contextual signals such as content attributes or creator relationships. As recommendation environments become increasingly multi-modal and metadata-rich, integrating sequential dynamics with structured representations has emerged as an important direction for improving robustness and expressiveness in user modeling.

2.3 Graph-based and Structure-Aware Recommendation

Recent advancements have shifted toward graph-based architectures to integrate auxiliary metadata and capture higher-order relationships among entities. Models such as Knowledge Graph Attention Networks (KGAT) [12] and Heterogeneous Graph Transformers (HGT) [3] demonstrate that relational inductive biases can significantly enhance representation quality in recommendation tasks. Specialized encoders including Relational Graph Convolutional Networks (R-GCN) [9] and Gated Graph Neural Networks (GGNN) [6] further highlight the importance of tailoring model structure to the topology of relational data.

More recent work has explored decomposing complex recommendation environments into multiple structural views, allowing models to capture complementary signals across interaction patterns, content relationships, and temporal dynamics. These approaches suggest that aligning model architecture with the inherent structure of the data can improve performance without requiring increasingly deep or monolithic networks.

Our work, HUG, builds on this perspective by introducing not only a graph based unified approach, but also a decomposed dual-view architecture that explicitly separates temporal interaction modeling from structural relationship modeling. Rather than relying on a single unified encoder, we match the inductive bias of each component to the statistical properties of its corresponding subgraph, enabling more efficient learning across heterogeneous signals while maintaining architectural flexibility.

3 Prediction Framework

Task Formalization: From CTR to CVR. While predicting an `is_click` event is conventionally treated as a Click-Through Rate (CTR) problem, we formalize our objective as Conversion Rate (CVR) prediction. In the short-video domain, the “click” action is functionally equivalent to a purchase conversion in traditional e-commerce. Due to the auto-play nature of modern feeds, simple exposure often guarantees a play; thus, the `is_click` signal in the KuaiRand dataset represents a conscious decision to engage deeply (analogous to a conversion) rather than a browsing selection.

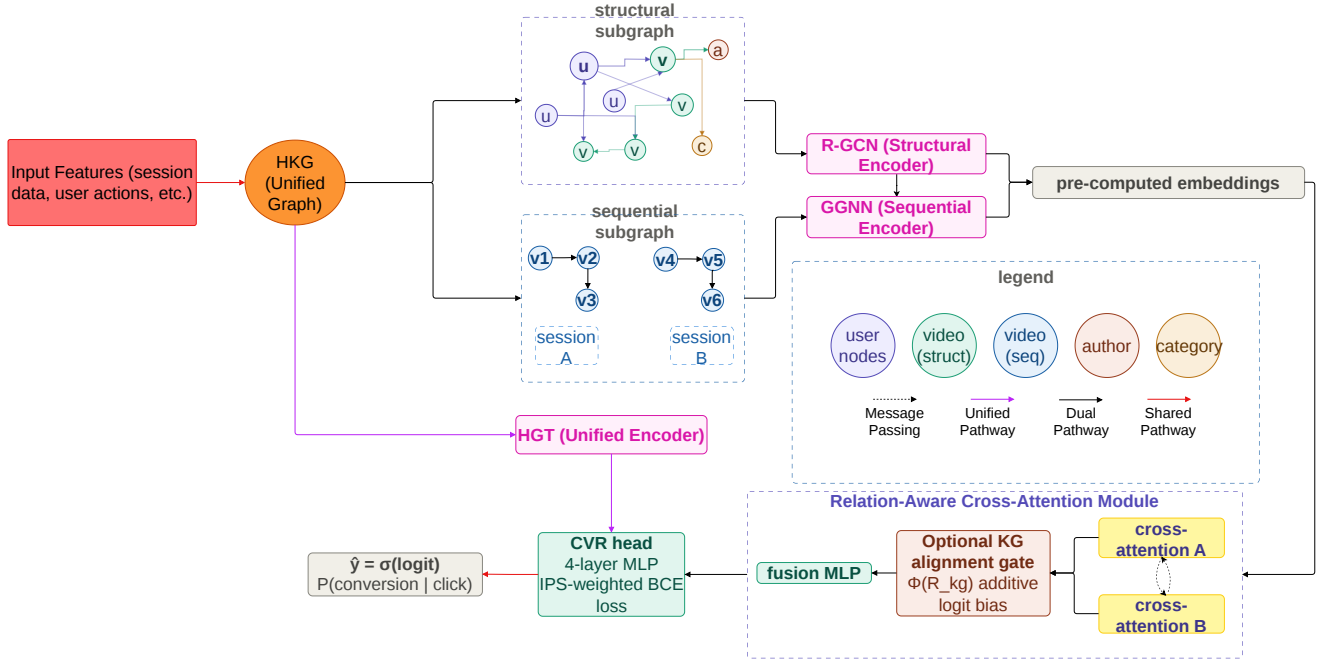


Figure 1: HUG Architecture.

Consequently, we model the probability $P(\text{conversion} | \text{click})$ as the primary objective, where the relational density of the HKG provides the necessary structural prior to overcome the extreme sparsity typically associated with these high-intent signals.

The HUG framework is built on the premise that effective CVR prediction requires the synthesis of two distinct signal modalities: **Structural Context** and **Sequential Intent**. To achieve this, we translate raw interaction logs and metadata into a high-fidelity graph substrate before applying our decomposed encoding pipeline. Our end-to-end architecture is shown in Figure 1.

3.1 Multi-Modal Graph Construction

From the input features and pre-computed sessions, we construct an HKG $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$. This graph serves as a unified latent substrate where relational attributes (e.g., user-follows-author) are elevated to first-class topological entities. We integrate five entity types (users, videos, sessions, authors, and categories) linked by twelve distinct interaction edges ranging from positive endorsements (likes, follows) to negative feedback (hate). This allows the framework to treat “browsing behavior” and “entity metadata” as a single, searchable manifold.

3.2 Decomposed Graph Encoding

While our unified approach operates over the global HKG $\mathcal{G}$, for the dual approach it is decomposed into two complementary subgraph views. Each subgraph view preserves the edge types most informative for its respective encoding objective.

The **structural subgraph** $\mathcal{G}_{str}$ retains four node types (users, videos, authors, and categories) connected by the seven relation types that encode static preference and content metadata (clicked,

liked, hated, commented, forwarded, made_by, and tagged_as). These edges capture long-term topological affinity between users and content, and between content and its semantic attributes.

The **sequential subgraph** $\mathcal{G}_{seq}$ retains only video nodes connected by directed next_in_session edges, derived from the interaction logs by sorting each user’s interactions by timestamp and placing a session boundary wherever the gap between consecutive interactions exceeds 30 minutes. Each directed edge carries the session identifier of the session it belongs to, enabling a GGNN to aggregate over temporally ordered video sequences and the session readout to pool per-session representations via soft attention with a recency gate. The video node feature tensor is shared between $\mathcal{G}_{str}$ and $\mathcal{G}_{seq}$, ensuring that content-side features inform both encoders without duplication.

To translate these specialized subgraphs into a unified latent space, we deploy a dual-encoding pipeline that matches the inductive bias of each architecture to the underlying graph topology. Within this pipeline, two GNNs serve as *structured feature extractors*: their role is to encode topological and temporal context into fixed embedding matrices. Task-specific learning (cross-view alignment and CVR prediction) occurs entirely within the alignment module and CVR head described in Section 3.4. The specific architectural configurations for these encoders are as follows:

- **StructuralGNN**: A 2-layer R-GCN over $\mathcal{G}_{str}$ that produces user embeddings $\mathbf{E}^u \in \mathbb{R}^{N_u \times d}$ and video embeddings $\mathbf{E}_{str}^v \in \mathbb{R}^{N_v \times d}$ by propagating relation-typed messages across users, videos, authors, and categories. An optional GAT refinement pass [10] over video-video co-engagement edges can be applied post-aggregation.

- **SequentialGNN:** A GGNN over $\mathcal{G}_{seq}$ that produces video embeddings $\mathbf{E}_{seq}^v \in \mathbb{R}^{N_v \times d}$ and session embeddings $\mathbf{E}^s \in \mathbb{R}^{N_s \times d}$ via GRU-style gated propagation along directed session edges, followed by a soft-attention readout with a learned recency gate that weights recent items more heavily.

3.3 HUG Variants

We propose two versions of the HUG framework with several customizable architectural settings:

- (1) **HUG-Unified:** This uses an HGT over the undivided HKG with direct concatenation fusion and no alignment module. This variant jointly encodes all relational and behavioral signals within a unified graph. Its primary advantage is architectural simplicity and full end-to-end relational message passing. However, by collapsing structural context (long-term preferences, metadata) and sequential dynamics (session-level intent) into a single encoder, it risks *signal conflation*, where temporally transient behaviors may interfere with stable preference representations. This variant allows us to isolate the benefit of heterogeneous graph modeling without explicit modality separation.
- (2) **HUG-Dual:** This utilizes the decomposed HKG and uses R-GCN and GGNN encoders with our cross-attention alignment module. Within this module is an optional KG alignment (KGA) gate and a recency gate. This variant introduces an explicit *dual-view decomposition*, separating structural and sequential signals into dedicated encoders (R-GCN for global relational structure and GGNN for session dynamics). This design aligns the inductive bias of each model with the nature of the signal, improving representation fidelity. The cross-attention module enables interaction between the two views; in the absence of the KG alignment gate, this interaction is driven purely by learned embedding similarity. With the presence of the KG alignment gate, KG paths provide a structural guide that biases cross-attention. The recency gate operates within the GGNN session readout: after soft-attention aggregates all items in a session into a summary embedding, a learned sigmoid gate interpolates per-dimension between this full-session summary and the last observed item. This allows the model to learn, per feature dimension, how much weight to place on long-term session history versus the most recently consumed item before the session embedding is passed to the alignment module.

3.4 Relation-aware Alignment Module

The alignment module fuses structural and sequential embeddings using a dual cross-attention mechanism. Let $h_{v,str}$ and h_s be the structural video and sequential session embeddings, respectively. For each attention head $k \in \{1, \dots, H\}$, $W_Q^{(k)}$, $W_K^{(k)}$, $W_V^{(k)} \in \mathbb{R}^{d \times d/H}$ are learned projection matrices that map embeddings into the query, key, and value spaces of head k .

The KG alignment gate $\Phi(R_{kg}) \in \mathbb{R}^H$ is a learned additive bias over the per-head attention logits, defined as:

$$\Phi(R_{kg}) = W_{kg} \mathbf{r}_{uv} \quad (1)$$

where $W_{kg} \in \mathbb{R}^{H \times d_{kg}}$ is a learned projection and $\mathbf{r}_{uv} \in \mathbb{R}^{d_{kg}}$ encodes the KG relation path features between user u and candidate item v . When a semantically relevant path exists in the HKG, for

example, a user $\rightarrow$ liked $\rightarrow$ video $\rightarrow$ tagged_as $\rightarrow$ category chain, this gate amplifies the corresponding attention heads, biasing the alignment toward graph-grounded structural signal.

The full alignment score for head k is then:

$$\alpha_k = \text{softmax} \left(\frac{(W_Q^{(k)} h_{v,str})^\top (W_K^{(k)} h_s)}{\sqrt{d/H}} + \Phi_k(R_{kg}) \right) \quad (2)$$

where $\Phi_k(R_{kg})$ denotes the k -th element of the gate vector, and d/H is the per-head dimension used for scaling. The gate is applied identically to both cross-attention directions, ensuring KG-derived structural context influences both the session-modulated video representation and the structure-modulated sequential representation.

The aligned representation z is then computed via a fusion Multi-Layer Perceptron (MLP):

$$z = \text{MLP}([\text{Align}(h_u, h_s) \parallel \text{Attn}(h_{v,str}, h_s) \parallel \text{Attn}(h_{v,seq}, h_{v,str})]) \quad (3)$$

where $\parallel$ denotes concatenation across the three components: (i) a user-session alignment, (ii) a structural-video-attended-by-session cross-attention, and (iii) a sequential-video-attended-by-structural-video cross-attention. This ensures the final representation is grounded in both temporal session intent and global relational context. The fused vector $z \in \mathbb{R}^d$ is passed to a CVR head consisting of a 4-layer MLP with BatchNorm and Dropout ($p = 0.2$) to predict the click probability $\hat{y} \in [0, 1]$. The effect of the KG alignment gate and recency gate is evaluated empirically in Section 5.

4 Experimental Design

To evaluate HUG, we implement DIN, BST, and KGAT as external sequence-based baselines. DIN and BST were implemented using the FuxiCTR v2.3.9 library [16, 17] while KGAT was adapted from the original work. We evaluate all models on the KuaiRand-1K dataset under identical data splits and optimization settings. Performance is measured using four standard ranking and classification metrics: Area Under the ROC Curve (AUC), Average Precision (AP), Binary Cross-Entropy (BCE), and Normalized Discounted Cumulative Gain (NDCG@10). AUC is the primary ranking metric, measuring global discrimination between positive and negative interactions. BCE measures calibration of the predicted probabilities and directly reflects the training objective. AP summarizes precision across recall thresholds and is particularly informative under KuaiRand's class imbalance. NDCG@10 evaluates the quality of the top-10 ranked items with a position discount. We compute NDCG@10 per user and macro-average across users rather than pooling all interactions globally, which mitigates the heavy-tailed interaction distribution in KuaiRand-1K where a small number of active users would otherwise dominate the score.

4.1 Dataset and Evaluation Protocol

All models are evaluated on KuaiRand-1K, which contains 11.7 million interactions across approximately 4.37 million unique videos and 1,000 users. We apply a temporal train/test split, assigning the earliest 80% of each user's interactions to training and the most recent 20% to test. This reflects the deployment setting in which a model trained on historical behavior must generalize to future interactions and prevents leakage of future engagement signals into

training. The positive rate across splits is approximately 37–39%, reflecting the active-user selection of the KuaiRand-1K subset.

4.2 Baselines

We compare against three external baselines that represent competitive industrial CVR and CTR modeling, encompassing both foundational sequence-based architectures and modern graph-based approaches.

- **DIN** [15]: Deep Interest Network applies target-aware attention over a user's video interaction history. It weights historical items by their relevance to the candidate item using a learned scoring function. We configure this with an embedding_dim of 16. The attention scoring MLP is defined with hidden layer widths $\mathcal{H}_{attn} = \{64\}$, where the set notation denotes the configurable sequence of layer sizes.
- **BST** [1]: Behavior Sequence Transformer captures sequential dependencies using a 2-layer transformer with positional embeddings. Unlike DIN's learned scoring MLP, BST employs a scaled dot-product mechanism. We set the embedding dimension to 16 and use $H = 4$ parallel attention heads to partition the embedding space.
- **KGAT** [12]: Knowledge Graph Attention Network models high-order relations by employing an attention-based message-passing mechanism over a collaborative knowledge graph. This serves as our primary graph-based baseline to isolate the benefit of the dual-view decomposition in HUG. We use 2 message-passing layers and an embedding dimension of 16.

The sequence-based baselines (DIN and BST) operate on flat interaction sequences and lack access to the multi-typed relational structure of the HKG. Their inclusion isolates the contribution of graph-based modeling; any performance gap between these and HUG-Unified reflects the value of heterogeneous relational encoding. Conversely, KGAT utilizes the graph structure but does not decouple structural and sequential views, allowing us to evaluate the specific impact of HUG-Dual's alignment-driven dual-view architecture. KGAT was re-implemented and adapted to operate on the same HKG as HUG, ensuring a fair graph-based comparison. These three models were selected as baselines due to their competitive performance, wide use, and open-source status. While some recent models, such as InterFormer [13], OneTrans [14], and HyFormer [4], may also be competitive, they are closed-source and thus cannot be tested. We make our model open-source to combat this growing issue.

4.3 Training Objective

A core challenge in CVR prediction is that standard interaction logs are collected under a production recommendation policy, resulting in a training distribution that reflects historical system behavior rather than the full space of possible impressions. In practice, this can lead models to overfit frequent exposure patterns and under-represent less commonly observed events.

Our training objective is based on the standard BCE loss commonly used for probabilistic classification in recommendation systems. To account for the mixed logging conditions present in the

Algorithm 1 HUG-Unified Training Per Epoch

Require: Unified heterogeneous graph $\mathcal{G}_{uni} = \mathcal{G}_{str} \cup \mathcal{G}_{seq}$, with node types $\mathcal{N} = \{\text{user, video, session}\}$ and edge types $\mathcal{R}$

Require: Training interactions $\mathcal{B} = \{(u_i, v_i, s_i, y_i, w_i)\}_{i=1}^N$

Require: Trainable parameters $\Theta_{hgt}, \Theta_{cvt}$

- 1: — **Stage 1: Encode the full graph once (no gradients)** —
- 2: $\mathbf{E}^u, \mathbf{E}^v, \mathbf{E}^s \leftarrow \text{HGT}(\mathcal{G}_{uni}; \Theta_{hgt})$ {joint user, video, and session embeddings via type-aware attention}
- 3: Store $\mathbf{E}^u, \mathbf{E}^v, \mathbf{E}^s$ in CPU memory
- 4: — **Stage 2: Train prediction head** —
- 5: **for** each mini-batch $b \subseteq \mathcal{B}$ **do**
- 6: Look up embeddings for each interaction in the batch:
- 7: $\mathbf{h}^u \leftarrow \mathbf{E}^u[u_b]$ {user embedding, shape $[B \times D]$ }
- 8: $\mathbf{h}^v \leftarrow \mathbf{E}^v[v_b]$ {video embedding (single, unified)}
- 9: $\mathbf{h}^s \leftarrow \mathbf{E}^s[s_b]$ {session embedding}
- 10: Concatenate and project (no cross-space alignment needed):
- 11: $\tilde{\mathbf{h}} \leftarrow \text{MLP}([\mathbf{h}^u \parallel \mathbf{h}^v \parallel \mathbf{h}^s]; \Theta_{cvt})$
- 12: Predict conversion probability:
- 13: $\hat{y}_b \leftarrow \sigma(\tilde{\mathbf{h}})$
- 14: Compute IPS-weighted loss and update trainable parameters:
- 15: $\mathcal{L} \leftarrow \frac{1}{B} \sum_{i \in b} \tilde{w}_i \cdot \text{BCE}(\hat{y}_i, y_i)$ { $\tilde{w}_i = \min(w_i, 10)$, normalised per batch}
- 16: $\Theta_{cvt} \leftarrow \text{Adam}(\nabla_{\Theta_{cvt}} \mathcal{L}; \text{grad clip} = 5.0)$
- 17: **end for**

KuaiRand dataset, we incorporate a lightweight importance weighting scheme that adjusts the contribution of samples originating from the dataset's random-policy intervention subset.

Specifically, KuaiRand includes a small fraction (0.37%) of recommendation slots filled by a uniform random policy [2]. Rather than treating these interactions identically to those generated by the production policy, we apply fixed scalar weights reflecting their relative frequency in the dataset. This weighting serves as a pragmatic normalization mechanism, ensuring that the rare random-policy samples are represented proportionally during optimization without introducing additional model complexity or requiring propensity estimation.

We assign $w = 1.0$ to random-policy interactions and $w = 0.9963$ to standard interactions, corresponding to the empirical replacement rate observed in the dataset. The resulting objective can be interpreted as a weighted variant of BCE, where importance weights modestly rescale each sample's contribution to the batch loss.

The IPS-weighted loss for a batch of size N is defined as:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \tilde{w}_i [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (4)$$

where $y_i \in \{0, 1\}$ denotes the binary click label, $\hat{y}_i$ is the predicted click probability, and $\tilde{w}_i$ are mean-normalized sample weights.

To maintain numerical stability during training, weights are clipped at a maximum value of 10.0 prior to normalization. This constraint bounds the effective contribution of any single sample while preserving the relative scaling between interaction types.

Algorithm 2 HUG-Dual Training Per Epoch

Require: Structural subgraph $\mathcal{G}_{str}$, sequential subgraph $\mathcal{G}_{seq}$

Require: Training interactions $\mathcal{B} = \{(u_i, v_i, s_i, y_i, w_i)\}_{i=1}^N$

Require: Frozen GNN parameters $\Theta_{str}, \Theta_{seq}$, and trainable parameters $\Theta_{aln}, \Theta_{cvr}$

- 1: — **Stage 1: Encode the full graph once (no gradients)** —
- 2: $\mathbf{E}^u, \mathbf{E}_{str}^v \leftarrow \text{R-GCN}(\mathcal{G}_{str}; \Theta_{str})$ {user and video embeddings from structural graph}
- 3: $\mathbf{E}_{seq}^v, \mathbf{E}^s \leftarrow \text{GGNN}(\mathcal{G}_{seq}; \Theta_{seq})$ {video and session embeddings from sequential graph}
- 4: Store $\mathbf{E}^u, \mathbf{E}_{str}^v, \mathbf{E}_{seq}^v, \mathbf{E}^s$ in CPU memory
- 5: — **Stage 2: Train alignment and prediction head** —
- 6: **for** each mini-batch $b \subseteq \mathcal{B}$ **do**
- 7: Look up embeddings for each interaction in the batch:
- 8: $\mathbf{h}^u \leftarrow \mathbf{E}^u[u_b]$ {user structural embedding, shape $[B \times D]$ }
- 9: $\mathbf{h}_{str}^v \leftarrow \mathbf{E}_{str}^v[v_b]$ {video structural embedding}
- 10: $\mathbf{h}_{seq}^v \leftarrow \mathbf{E}_{seq}^v[v_b]$ {video sequential embedding}
- 11: $\mathbf{h}^s \leftarrow \mathbf{E}^s[s_b]$ {session embedding}
- 12: Fuse the four embeddings with cross-attention:
- 13: $\tilde{\mathbf{h}} \leftarrow \text{AlignmentModule}(\mathbf{h}^u, \mathbf{h}_{str}^v, \mathbf{h}_{seq}^v, \mathbf{h}^s; \Theta_{aln})$
- 14: Predict conversion probability:
- 15: $\hat{y}_b \leftarrow \sigma(\text{MLP}(\tilde{\mathbf{h}}; \Theta_{cvr}))$
- 16: Compute IPS-weighted loss and update trainable parameters:
- 17: $\mathcal{L} \leftarrow \frac{1}{B} \sum_{i \in b} \hat{w}_i \cdot \text{BCE}(\hat{y}_i, y_i)$ { $\hat{w}_i = \min(w_i, 10)$, normalised per batch}
- 18: $\Theta_{aln}, \Theta_{cvr} \leftarrow \text{Adam}(\nabla_{\Theta_{aln}, \Theta_{cvr}} \mathcal{L}; \text{grad clip} = 5.0)$
- 19: **end for**

4.4 Implementation

All models are trained using the Adam [5] optimiser ($\beta_1 = 0.9, \beta_2 = 0.999, \text{lr}=10^{-3}, \text{weight_decay}=10^{-5}$). HUG variants additionally apply CosineAnnealingLR [7] scheduling with $\eta_{min} = 10^{-4}$ over $T_{max} = 20$ epochs, decaying the learning rate smoothly to prevent oscillation in late training. DIN and BST are implemented with identical optimiser settings to ensure fair comparison. KGAT has been adapted for the KuaiRand dataset with the same settings. All models are trained for 20 epochs with a batch size of 2048 and mixed-precision training (AMP) enabled.

The HUG-Dual architecture uses a GNN hidden dimension of $d = 128$ and output embedding dimension of $D = 64$. Node type input projections map each entity's raw feature vector to the shared hidden dimension. The R-GCN structural encoder and the GGNN sequential encoder both apply 2 message-passing steps. KGAT is configured with 2 attention-based message-passing layers and an embedding_dim of 16 to match the sequence-based baselines.

The alignment module uses $H = 4$ attention heads, giving a per-head dimension of $D/H = 16$. The fusion MLP consists of two linear layers with an intermediate dimension of $2D = 128$, ReLU activation, and Dropout ($p = 0.1$). The CVR head for all HUG variants is a 4-layer MLP with hidden dimensions $\{D, D/2\}$, BatchNorm, and Dropout ($p = 0.2$). Gradient norms are clipped to 5.0 during the alignment and head update steps. Algorithm 1 and Algorithm 2 detail the training steps per epoch for HUG-Unified and HUG-Dual

Table 2: Hyperparameter configurations for all models. Dashes (–) indicate settings not applicable to a given model.

Hyperparameter	DIN	BST	KGAT	HUG-Unified	HUG-Dual
<i>Optimiser & Training</i>					
Optimiser	Adam	Adam	Adam	Adam	Adam
Learning rate	10^{-3}	10^{-3}	10^{-3}	10^{-3}	10^{-3}
Weight decay	10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}
LR schedule	–	–	–	Cosine	Cosine
Batch size	2048	2048	2048	2048	2048
Epochs	20	20	20	20	20
Gradient clip	–	–	–	5.0	5.0
<i>Architecture</i>					
Embedding dim	16	16	16	–	–
Attention dim	64	–	–	–	–
Transformer heads	–	4	–	–	–
GNN hidden dim d	–	–	–	128	128
Output dim D	–	–	–	64	64
GNN/KGAT layers	–	–	2	2	2
Alignment heads H	–	–	–	–	4
Fusion MLP hidden	–	–	–	–	128
<i>Regularisation & Head</i>					
Dropout	–	–	0.1	0.2	0.2/0.1
BatchNorm	–	–	✓	✓	✓
CVR head layers	–	–	–	4	4

respectively. Table 2 summarizes the hyperparameter settings for training and the optimiser.

5 Results

The test performance of all models on the KuaiRand-1K dataset along with their hyperparameter settings is summarized in Table 3. We evaluate each model on its ability to predict conversion events in the presence of selection bias and signal sparsity.

5.1 Comparative Analysis

HUG-Dual achieves AUC = 0.7626 and AP = 0.6510 (Figure 2), representing an **8.81% relative improvement in AUC** and a **19.60% improvement in AP** over the best sequence baseline (BST; AUC = 0.7009, AP = 0.5443). Note that Figure 2 does not take into account hyperparameter ablation of L, d, h . **HUG-Dual (KGA On)** achieves essentially identical results (AUC = 0.7625, AP = 0.6509), differing by less than 0.02% on every metric; we treat the two as equivalent in the analysis below. The gain in AP is particularly significant for high-sparsity CVR tasks, where precision at the top of the ranked list is paramount. We also compare against KGAT [12] trained under identical conditions. At final epoch 20, KGAT (IPS) achieves AUC = 0.6886, below both sequence baselines.

Moving from sequence models to **HUG-Unified** raises AUC from 0.7009 to 0.7306 (+4.24%) and AP from 0.5443 to 0.6036 (+10.90%), confirming that elevating entity metadata and diverse interaction types to topological features provides a richer signal than flat sequences. Notably, HUG-Unified without IPS (AUC = 0.7514) outperforms its IPS-weighted counterpart (AUC = 0.7306).

Moving from **HUG-Unified** to **HUG-Dual** yields further gains: AUC improves from 0.7306 to 0.7626 (+4.38%) and AP from 0.6036 to 0.6510 (+7.85%), with BCE loss falling from 0.5803 to 0.5533. Separating structural and sequential signals into dedicated encoders,

Table 3: Complete registry of all experimental runs. KGA: knowledge graph alignment embedding dimension (0 = disabled). IPS: inverse propensity score weighting. RG: recency gate. L : GNN layers; h/d : hidden/output dimension. Metrics are from the final test epoch. Dashes (–) indicate settings not applicable to the given model. Best result per column is bolded.

Run	KGA	IPS	RG	L	h	d	AUC $\uparrow$	AP $\uparrow$	BCE $\downarrow$	NDCG@10 $\uparrow$
DIN [15]	–	–	–	–	–	16	0.6903	0.5392	0.6000	0.7065
BST [1]	–	–	–	2	–	16	0.7009	0.5443	0.5924	0.8553
KGAT (no IPS) [12]	64		–	2	128	64	0.6877	0.5414	0.6097	0.4862
KGAT (IPS) [12]	64	✓	–	2	128	64	0.6886	0.5446	0.6079	0.4813
HUG-Unified (no IPS)	0		✓	2	128	64	0.7514	0.6312	0.5638	0.7784
HUG-Unified	0	✓	✓	2	128	64	0.7306	0.6036	0.5803	0.8003
HUG-Dual (KGA off)	0	✓	✓	2	128	64	0.7626	0.6510	0.5533	0.8554
HUG-Dual (KGA on)	64	✓	✓	2	128	64	0.7625	0.6509	0.5560	0.8474
HUG-Dual (no IPS)	64		✓	2	128	64	0.7622	0.6510	0.5578	0.8484
HUG-Dual (no RG)	64	✓		2	128	64	0.7670	0.6567	0.5497	0.8514
HUG-Dual (no IPS, no RG)	64			2	128	64	0.7676	0.6575	0.5490	0.8535
HUG-Dual ($L=1, h=128, d=64$)	64	✓	✓	1	128	64	0.7683	0.6578	0.5486	0.8339
HUG-Dual ($L=3, h=128, d=64$)	64	✓	✓	3	128	64	0.7549	0.6379	0.5603	0.8543
HUG-Dual ($L=2, h=64, d=32$)	32	✓	✓	2	64	32	0.7562	0.6399	0.5657	0.8229
HUG-Dual ($L=2, h=256, d=128$)	128	✓	✓	2	256	128	0.7669	0.6567	0.5541	0.8548

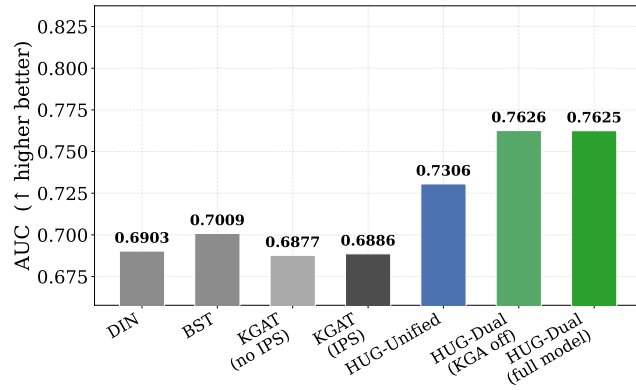


Figure 2: AUC on KuaiRand-1K across the model progression. Full metric results in Table 3.

each matched to its data modality, is the primary driver of HUG's performance advantage.

5.2 Ablation Study

Ablation results across KGA, RG, and IPS activation are shown in Table 3. Activating the KG alignment gate (**KGA On**) yields results statistically indistinguishable from **KGA Off** (AUC = 0.7625 vs. 0.7626, AP = 0.6509 vs. 0.6510). The alignment gate remains architecturally motivated and may provide clearer gains on graphs with denser

or more varied relational paths than KuaiRand-1K. IPS weighting has negligible effect (AUC change -0.0003): KuaiRand-1K's random-policy subset covers only 0.37% of impressions, making weights near-uniform ($w^{\text{biased}}=0.9963$), so re-weighting produces no measurable shift in gradient updates. Removing the recency gate consistently improves all metrics (AUC $+0.0045$, AP $+0.0058$, BCE -0.0063). We attribute this to the GGNN already encoding temporal order through directed session edges and GRU-style gating, so the explicit last-item blending layer is redundant and overfits. Removing both components together gives the best results across all four metrics, confirming the effects are independent and additive. Both components remain theoretically motivated for settings with stronger exposure bias or longer sessions. HUG-Unified serves as a stepping-stone in the model-progression analysis (Section 5.1) rather than a standalone candidate for component-level ablation.

5.3 Hyperparameter Sensitivity

As shown in Table 3, a single GNN layer ($L=1$) outperforms the default $L=2$ on all metrics. A second propagation step aggregates 2-hop neighbours that introduce over-smoothing; three layers ($L=3$) amplifies this further. The one trade-off is NDCG@10, where $L=2$ (0.8548) and $L=3$ (0.8543) exceed $L=1$ (0.8339), suggesting deeper propagation refines top-rank ordering while hurting global precision. The smaller 2 layer configuration ($h=64, d=32$) underperforms on most metrics, while the larger ($h=256, d=128$) offers modest gains ($+0.0044$ AUC) with diminishing returns. The combination

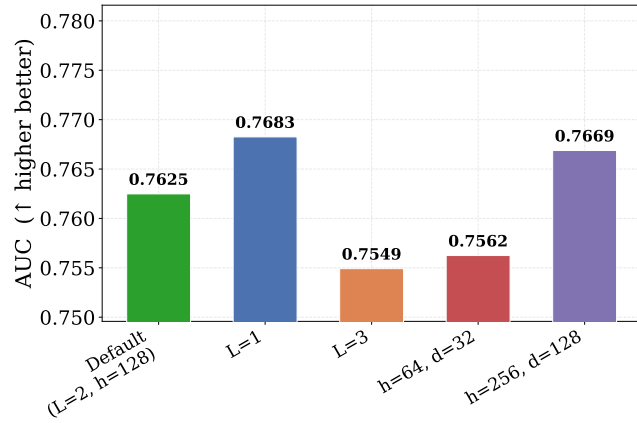


Figure 3: AUC under hyperparameter variations (HUG-Dual, IPS on, RG on). $L=1$ is the best configuration for AUC and AP; full metric results in Table 3.

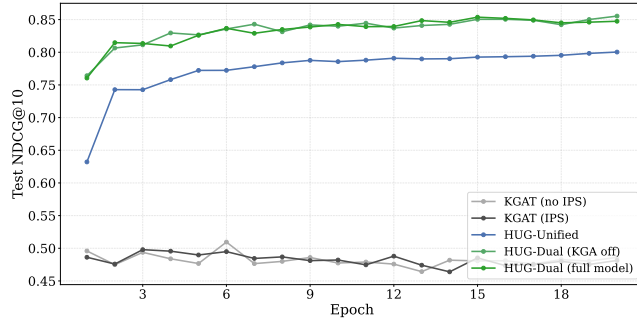


Figure 4: NDCG@10 over 20 training epochs.

$L=1$, $h=128$, $d=64$ is a stable operating point balancing both objectives. Figure 3 displays AUC’s sensitivity to hyperparameter ablation.

5.4 Convergence and Ranking Stability

Figure 4 shows that KGAT has a fairly stable NDCG@10 across epochs and has some periods of gain followed by loss. This is a sharp contrast with all HUG variants which improve smoothly across all 20 epochs under cosine LR scheduling. HUG-Dual (KGA Off) reaches the highest final NDCG@10 of 0.8554, matching BST (0.8553) on top-rank quality while outperforming it by 8.81% on AUC and 19.60% on AP, demonstrating that the dual-view architecture achieves ranking parity with the best sequential model while capturing the relational structure needed for superior precision across the full ranking.

6 Conclusion

In this work, we presented HUG, a modular framework for conversion rate prediction in short-video recommendation. By unifying disparate behavioral and metadata signals into an HKG, we addressed the fundamental challenges of signal sparsity and relational complexity. Our primary architectural contributions (the

decomposition of the HKG into structural and sequential views, dual encoding approach, and cross attention module) demonstrates that matching encoder inductive biases (R-GCN and GGNN) to the temporal nature of the data significantly outperforms both traditional sequence models and a graph-based KGAT baseline.

Our empirical evaluation on KuaiRand-1K validates this approach. HUG-Dual achieves an AUC of 0.7683 and an AP of 0.6578, marking an 8.81% and 19.60% relative improvement respectively over BST. Sensitivity analysis reveals that a single GNN layer ($L=1$) maximises AUC (0.7683), suggesting that on this dataset scale deeper message passing introduces more noise than signal. Ablation results show that IPS weighting and the recency gate do not hurt performance and remain architecturally motivated for settings with stronger exposure bias; however, their marginal impact on KuaiRand-1K, where propensity weights differ from unity by only 0.37%, calls for evaluation on datasets with a larger random-policy intervention fraction.

Several promising avenues for future work remain. First, incorporating hierarchical session-graph attention could further capture cross-session intent evolution. Second, the current alignment gate relies on pre-defined KG paths; automated meta-path discovery or reinforcement learning-based path sampling could improve adaptability to new domains. Third, a combined sensitivity sweep over L and h (e.g., $L=1$, $h=256$) may yield further gains not captured by the single-dimension experiments conducted here.

7 Supplementary Materials

Our code is available at <https://anonymous.4open.science/r/GCIC-3B57>. This repository also includes several figures highlighting our other metrics, such as training convergence of AUC, AP, BCE, and NDCG@10 across epochs. The KuaiRand dataset is available at <https://kuairand.com>.

8 Acknowledgments

Portions of the source code for this work were generated or refined using generative artificial intelligence tools. The authors remain solely responsible for the final implementation and any errors or omissions therein.

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